

Final Project Report

Introduction

This research is motivated by coming up with a linear regression model for predicting a dependent variable of the current dataset considering the many independent variables because determining the age of abalone—sea creatures used as cuisines—is essential in identifying abalones that are usable, non-usable, and harmful. The age of abalone is determined by cutting the shell through the cone, staining it, and counting the number of rings through a microscope, which is a boring and time-consuming task. Therefore, the dataset is interesting and challenges the researcher to look for inter-relevant variables and explore best prediction models, including linear regression model. The current dataset entails 4177 records and illustrates some 9 characteristics of these observations. The dataset collects information on the sex, diameter, weight, height, length, rings, shell weight, shucked weight, whole weight, and viscera weight of abalone. The preliminary goal was to develop a prediction model and use 70% of the data for training and 30% for testing, but it didn't give an appropriate prediction model.

Method

Linear Regression Model has been used for the current dataset because of its continuous values. For obtaining the linear regression model we follow the below steps:

- a. We carry out some data pre-processing procedures to make sure the data is clean
- b. We visualize our data and change the categorical variables into discrete variables
- c. We use 80% of data for training and use 20% of it to test the prediction model
- d. We visualize our prediction model and evaluate our results
- e. We calculate RMSE and MSE

- f. We check the accuracy rate of our model

Discussion and Results

This research paper aimed to come up with a linear regression model and predict some of the continuous variables such as diameter and length. The first linear regression model developed in this research had some 97% accuracy which is an ideal prediction model and can be used for further analysis. On the other hand, the second linear regression model had a 83% accuracy, which is not an ideal prediction model to be used. To predict continuous variables, the study attempts to reduce the Means Square Error (MSE)—the difference between the actual and the predicted points. The study also used 80% for training and 20% of the data to test the prediction model.

Conclusion

The study found that in linear regression model the more MSE value get smaller the more reliable the prediction model become. Also, lower MSE values indicate that the prediction model has a high level of accuracy. Moreover, when more than two variables are studied, their relationship visualized, and their prediction model develop, their accuracy rate decreases and their MSE and RMSE increases. However, the present project has some limitations. One of the shortcomings of this study is that it cannot portray the relationship between more than three continuous variables. In other words, it is not possible to plot the scatter plot and linear regression line of these variables because their size don't fit. Therefore, they cannot be visualized and their relationship explored. Nevertheless, if given more time, the research could improve the questions being asked, develop better prediction models with better accuracy rate, and examine them in more details using data visualization, linear regression model, and data preprocessing techniques from online sources—YouTube videos, tech websites, and data scientists' personal blogs.

References

Kaggle. (2022). Abalone Dataset. Retrieved from <https://www.kaggle.com/datasets/rodolfomendes/abalone-dataset>

Machine Learning Repository. (2022). Abalone Data Set. Retrieved from <https://archive.ics.uci.edu/ml/datasets/abalone>